

50 g Lab Cup · BX-01

BESTNEW

Closed-cell pour is **50.00 g B (includes 5.00 g HFO-1233zd(E)) + 41.40 g A**. HFO is mandatory. Cream/gel are sped up. Mix 4–5 s. Weigh to 0.01 g.

Cup size	
B-side (HFO in)	50.00 g
of which HFO-1233zd(E)	5.00 g
A-side (PMDI 31 % NCO)	41.40 g
Total	91.40 g
Index	125
Mix	A into B · 4–5 s · 2 500–3 000 rpm
B cup temperature	15–18 °C after HFO

Trial header	
Date	
Operator	
Drum · high-OH	TB-307 / PE 104
Drum · flow	TEROL 250 / PE 103
Cup T	A 21–23 °C · B 15–18 °C
Cup ID	BX-01-50-

BX-01 weigh sheet · 50.00 g B · HFO mandatory

Weigh in this order. Tare a dry, closable B cup. Water before catalysts. **HFO last**, cup at 15–18 °C, cap at once. Write the actual in the last column.

#	INGREDIENT	TARGET G	ACTUAL G
1	ISOEXTER TB-307 / KIMpol PE 104	20.65	
2	TEROL 250 / KIMpol PE 103	12.40	
3	CREAPOL PD 230	2.00	
4	TCPP	3.00	
5	TEP	2.00	
6	Reactive FR	1.00	
7	Emulsifier	0.875	
8	Closed-cell silicone	0.75	
9	Water (deionized / distilled)	0.60	
10	PC-15	0.725	
11	DABCO 2040	0.550	
12	CAT-T	0.275	
13	PC-37	0.175	
14	HFO-1233zd(E) — last, cold, cap	5.00	
	B-side total	50.000	
A	PMDI, 31.0 % NCO	41.40	

HFO 5.00 g is 10 pbw on 50 g. Water 0.60 g is 1.20 pbw. Water 0.60 ÷ 9.0 = 0.0667 eq → 11.29 g PMDI at index 125. Polyols (TB-307 20.65 + TEROL 12.40 + PD 230 2.00) → 30.11 g PMDI at index 125. Total A = **41.40 g**. Do not pad a short cup with extra water. If HFO flashes off while weighing, recap and reweigh — do not add extra water to compensate.

Checklist · before the pour

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Safety and bench	
☐	Gloves, goggles, lab coat
☐	Fume hood or good ventilation
☐	Balance reads 0.01 g and is level
☐	Two dry cups, labeled A and B
☐	Timer ready (cream / gel / rise / tack-free)
☐	Thermometer · A at 21–23 °C · B at 15–18 °C after HFO
☐	Mixer 2 500–3 000 rpm, clean blade
☐	Paper towel / waste cup for the blade

Drums and grades	
☐	High-OH drum is TB-307 or PE 104 — write which
☐	Flow drum is TEROL 250 or PE 103 — write which
☐	PD 230, TCPP, TEP, reactive FR on the bench
☐	Closed-cell silicone — not the open-cell grade
☐	PC-15, PC-37, CAT-T, DABCO 2040
☐	PMDI is 31.0 % NCO (write lot / %NCO)
☐	HFO-1233zd(E) on the bench — mandatory for this cup
☐	HFO cylinder / bottle cold · hood on · no open flame
☐	Closable B cup ready (HFO boils at ~19 °C)
☐	No sucrose, Mannich, K-15, DMCHA or pentane in this cup

Checklist · pour and record

Pour	
☐	Resin mixed homogeneous, then chilled to 15–18 °C
☐	HFO 5.00 g weighed last · cup capped at once
☐	A weighed 41.40 g in a dry cup
☐	Pour A into B — start the clock at mix start
☐	Mix 4–5 s only — cream is 3–6 s, then do not touch
☐	Leave cup 24 h before cutting

Record (write)	
Cream	s target 3–6
Gel	s target 8–15
Rise	s
Tack-free	s target 15–28
Collapse / shrink / scorch	
Free-rise density	kg/m³ target 32–38
Pass?	3 cups, no collapse, times repeat

Half-cup (25 g B, 2.50 g HFO, 20.70 g A) is allowed only if the 0.175 g PC-37 weigh is trusted. Prefer 50.00 g B. Mix 4–5 s. Open-cell 50 g is the next sheet.

50 g Lab Cup · BX-02 Open Cell

BESTNEW

Open-cell 50 g cup. **No HFO** on this pour. Only if sucrose is on the floor. Do not blend with BX-01 or OC-01.

Cup size	
B-side	50.00 g
A-side (PMDI 31 % NCO)	65.65 g at index 105
Total	115.65 g
Index	100–110 (calc. 105)
Mix	A into B · 4–5 s · 2 500–3 000 rpm
Cup temperature	A and B 21–23 °C

Trial header	
Date	
Operator	
Sucrose grade / OH#	assumed OH 400 for A-side
Flow polyester	TEROL 250 / PE 103
Cup T (A / B)	21–23 °C
Cup ID	BX-02-50-

BX-02 weigh sheet · 50.00 g B

Weigh in this order. Tare a dry B cup. Water before catalysts. Open-cell silicone only. No HFO. A-side assumes sucrose OH 400 — re-index when the lot OH is known.

#	INGREDIENT	TARGET G	ACTUAL G
1	Sucrose / sucrose polyol	19.00	
2	TEROL 250 / KIMpol PE 103	8.00	
3	TEP	10.45	
4	Reactive FR	3.50	
5	Ecosurf SA-9	2.00	
6	Open-cell silicone	0.75	
7	Water (deionized / distilled)	2.60	
8	PC-37	0.85	
9	CAT-T	0.80	
10	PC-15	1.40	
11	BDMAEE or DEDEE	0.45	
12	Biocide	0.20	
	B-side total	50.00	
A	PMDI, 31.0 % NCO	65.65	

Water 2.60 g is 5.2 pbw. Water 2.60 ÷ 9.0 = 0.2889 eq → 41.10 g PMDI at index 105. Sucrose 19.00 (OH 400) + TEROL 8.00 (OH 260) → 24.55 g PMDI at index 105. Total A = **65.65 g**. If sucrose OH is 350, A ≈ 62 g. If 450, A ≈ 70 g. Do not force 1:1. No HFO on this cup.

Checklist · open-cell 50 g

NEW

Before	
☐	Gloves, goggles, lab coat
☐	Balance 0.01 g · two dry cups A and B
☐	A and B at 21–23 °C
☐	Sucrose polyol on the floor — write grade / OH#
☐	TEROL 250 or PE 103 — write which
☐	Open-cell silicone — not the closed-cell grade
☐	PC-37, CAT-T, PC-15, BDMAEE, biocide
☐	No HFO / pentane on this cup

Pour and record	
☐	B mixed homogeneous
☐	A weighed 65.65 g (or re-indexed)
☐	Pour A into B · mix 4–5 s · clock from mix start
☐	Do not touch after cream · cut at 24 h
Cream	s target 2–4
Rise	s
Tack-free	s
Free-rise density	kg/m³ target 16–22
Cells	open / tight / collapse

OC-01 memo screen · 50.00 g B

High-water System C. Not the best card. No HFO. A-side is calculated: 12.00 g water + 16.50 g sucrose (OH 400) against PMDI 31.0 % NCO (EW 135.5).

INGREDIENT	G	INGREDIENT	G
Sucrose / sucrose polyol	16.50	Reactive FR	4.00
TEP	8.50	Ecosurf SA-9	3.00
Open-cell silicone	0.60	Water	12.00
PC-37	1.20	CAT-T	1.20
PC-15	2.10	BDMAEE / biocide	0.50 / 0.40
B-side	50.00	A-side PMDI, index 100	196.6

Precise A-side	
Water 12.00 ÷ 9.0 = 1.3333 eq × 135.5 = 180.67 g PMDI	
Sucrose 16.50 · OH 400 · EW 56 100/400 = 140.25 → 16.50 ÷ 140.25 = 0.1176 eq × 135.5 = 15.94 g PMDI	
Index 100: 180.67 + 15.94 = 196.6 g A . Index 90 = 176.9 g · index 105 = 206.4 g. Do not pour 20–28 g A — that never pays the water.	

If cells stay closed on BX-02: +0.20 g water or −0.10 g PC-15. If it collapses: +0.15 g PC-15. Do not pour BX-02 and OC-01 as one blend.